

Understanding Vector Subspaces

Lecture Notes

Zamir Hussain
Department of Mathematics
University of Wah

November 18, 2025

Contents

1	The Core Idea: What is a Subspace?	2
2	A Powerful Shortcut: The Span Theorem	2
3	Worked Examples	3
3.1	Example 1: A Subspace Defined by Span	3
3.2	Example 2: A Set Failing All Three Properties	3
3.3	Example 3: A Set Defined by an Inequality	4
4	Summary & Applications	5
4.1	Summary of Key Concepts	5
5	Exercise Problems	5

Introduction

In linear algebra, a **vector space** is a collection of objects called vectors that can be added together and multiplied by scalars, following a set of ten axioms. A **subspace** is a special subset of a larger vector space that is, in itself, a complete vector space.

This lecture explores how to determine if a given subset of vectors, U , from a known vector space (like $\mathbb{R}^2$ or $\mathbb{R}^3$) is a subspace. We will cover two powerful methods for verification:

1. The **Span Theorem**, a quick and elegant method for sets defined by linear combinations.
2. The **Three-Property Subspace Test**, a direct check of the essential conditions for any subset.

1 The Core Idea: What is a Subspace?

A subset U of a vector space V is a **subspace** of V if U itself is a vector space under the same operations of vector addition and scalar multiplication defined on V .

Instead of checking all ten vector space axioms, we can use a simplified test. If we know U is a subset of a vector space V , we only need to verify three critical properties.

💡 The Three-Property Subspace Test

A subset U of a vector space V is a subspace of V if and only if it satisfies the following three conditions:

1. **Contains the Zero Vector:** The zero vector of V must be in U .
2. **Closed Under Addition:** For any two vectors $\mathbf{u}$ and $\mathbf{v}$ in U , their sum $\mathbf{u} + \mathbf{v}$ must also be in U .
3. **Closed Under Scalar Multiplication:** For any vector $\mathbf{u}$ in U and any scalar c , the product $c\mathbf{u}$ must also be in U .

If a subset fails even one of these conditions, it is **not** a subspace.

2 A Powerful Shortcut: The Span Theorem

A common way to define a subset of vectors is as all possible linear combinations of a given set of vectors. This leads to a very useful theorem.

💡 The Span Theorem

If $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ are vectors in a vector space V , then the set of all their possible linear combinations, known as the **span** of these vectors, is always a subspace of V .

$\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ is a subspace of V .

The span is the collection of all vectors that can be written in the form $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_p\mathbf{v}_p$, where $c_1, c_2, \dots$ are any scalars.

3 Worked Examples

Let's apply these concepts to determine if different subsets are subspaces.

3.1 Example 1: A Subspace Defined by Span

Problem: Let U be the set of all vectors in $\mathbb{R}^3$ of the form $\begin{bmatrix} 2r - s \\ 3r \\ r + s \end{bmatrix}$, where r and s are any real numbers. Is U a subspace of $\mathbb{R}^3$?

Solution (Step-by-Step using the Span Theorem)

1. **Decompose the General Vector:** Separate the terms involving r from the terms involving s .

$$\begin{bmatrix} 2r - s \\ 3r \\ r + s \end{bmatrix} = \begin{bmatrix} 2r \\ 3r \\ r \end{bmatrix} + \begin{bmatrix} -s \\ 0 \\ s \end{bmatrix}$$

2. **Factor out the Scalars:** Factor r from the first vector and s from the second.

$$= r \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} + s \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

3. **Identify the Underlying Vectors:** This shows that every vector in U is a linear combination of two specific vectors: $\mathbf{v}_1 = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$.

4. **Connect to the Span Definition:** By definition, the set of all such linear combinations is the span of these two vectors.

$$U = \text{Span} \left\{ \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

5. **Apply the Span Theorem:** We identified two vectors, $\mathbf{v}_1$ and $\mathbf{v}_2$, which are in $\mathbb{R}^3$. We have shown that U is their span. The theorem states that the span of vectors from a vector space is always a subspace of that space.

Conclusion: By the Span Theorem, U is a subspace of $\mathbb{R}^3$.

3.2 Example 2: A Set Failing All Three Properties

Problem: Let U be the set of all vectors in $\mathbb{R}^3$ of the form $\begin{bmatrix} 2r - s \\ 2 \\ r + s \end{bmatrix}$, where r and s are any real numbers. Is U a subspace of $\mathbb{R}^3$?

Solution (Step-by-Step using the Subspace Test)

We cannot easily write this as a span because of the constant 2. We must use the three-property test.

1. **Check for the Zero Vector:** The zero vector in $\mathbb{R}^3$ is $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$. For a vector in U , the second component is *always* 2. It is impossible for this component to be 0. Therefore, the zero vector is **not in** U .

Conclusion: Since U fails the first test, we can immediately conclude that U is **not a subspace of** $\mathbb{R}^3$. (We check the others for practice.)

2. **Check for Closure Under Addition:** Let $\mathbf{u} = \begin{bmatrix} 2r_1 - s_1 \\ 2 \\ r_1 + s_1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 2r_2 - s_2 \\ 2 \\ r_2 + s_2 \end{bmatrix}$ be two vectors in U .

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} 2(r_1 + r_2) - (s_1 + s_2) \\ 4 \\ (r_1 + r_2) + (s_1 + s_2) \end{bmatrix}$$

The second component of the sum is 4, not 2. Therefore, $\mathbf{u} + \mathbf{v}$ is **not in** U .

3. **Check for Closure Under Scalar Multiplication:** Let $\mathbf{u} = \begin{bmatrix} 2r - s \\ 2 \\ r + s \end{bmatrix}$ be in U and let c be a scalar.

$$c\mathbf{u} = \begin{bmatrix} c(2r - s) \\ 2c \\ c(r + s) \end{bmatrix}$$

The second component is $2c$. This is only equal to 2 if $c = 1$. For any other scalar, the result is not in U . Thus, U is not closed under scalar multiplication.

3.3 Example 3: A Set Defined by an Inequality

Problem: Let U be the set of all vectors $\begin{bmatrix} x \\ y \end{bmatrix}$ in $\mathbb{R}^2$ such that $x > 1$. Is U a subspace of $\mathbb{R}^2$?

Solution (Step-by-Step using the Subspace Test)

1. **Check for the Zero Vector:** The zero vector is $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$. Here, $x = 0$. The condition for being in U is $x > 1$. Since $0 \not> 1$, the zero vector is **not in** U .

Conclusion: U is not a subspace of $\mathbb{R}^2$.

2. **Check for Closure Under Scalar Multiplication (Counterexample):** We can also show failure with a counterexample.

- Choose a vector in U , e.g., $\mathbf{u} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$ (since $2 > 1$).
- Choose a scalar, e.g., $c = -1$.

- Calculate the product: $c\mathbf{u} = -1 \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$.
- Check if the result is in U . The first component is -2 . Since $-2 \not\geq 1$, the result is **not in** U .

This single counterexample proves that U is not closed under scalar multiplication.

4 Summary & Applications

4.1 Summary of Key Concepts

- A **subspace** is a subset of a vector space that is itself a vector space.
- To verify if a subset U is a subspace, you must check three conditions: **(1)** it contains the zero vector, **(2)** it is closed under addition, and **(3)** it is closed under scalar multiplication.
- If a set U can be described as the **span** of a collection of vectors from V , then U is automatically a subspace of V .
- A single failure (a counterexample) for any of the three properties is enough to prove a set is **not** a subspace.

Practical Applications in Data Science & ML

- **Eigenspaces:** For a given matrix (e.g., a covariance matrix), the set of all eigenvectors corresponding to a single eigenvalue, plus the zero vector, forms a subspace called an **eigenspace**. Dimensionality reduction techniques operate within these subspaces.
- **Solution Spaces (Null Space):** The set of all solutions to a homogeneous system ' $Ax = 0$ ' is the **null space** of A . The null space is always a subspace. This is crucial in understanding models like linear regression, especially with multicollinearity.
- **Feature Spaces and Dimensionality Reduction:** Techniques like PCA work by projecting high-dimensional data onto a lower-dimensional **subspace**. This subspace is spanned by the principal components that capture the most variance in the data.

5 Exercise Problems

1. Let U be the set of all vectors $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ in $\mathbb{R}^3$ such that $x + y - z = 0$. Is U a subspace of $\mathbb{R}^3$? (Hint: Can you write a general vector in U as a span?)
2. Let W be the set of all vectors $\begin{bmatrix} x \\ y \end{bmatrix}$ in $\mathbb{R}^2$ such that $x^2 - y = 0$. Is W a subspace of $\mathbb{R}^2$? Check the three properties.

3. Let S be the set of all 2×2 matrices A such that $\det(A) = 0$. Is S a subspace of the vector space of all 2×2 matrices? (Hint: Check closure under addition with a counterexample.)
4. True or False: The set of all vectors $\begin{bmatrix} x \\ y \end{bmatrix}$ in $\mathbb{R}^2$ such that $x \geq 0$ and $y \geq 0$ (the first quadrant) is a subspace of $\mathbb{R}^2$. Justify your answer.